

FinTrust Digital Banking — Week 2: Data Preparation, EDA & Baseline Modeling

Project Overview & Week 2 Objectives

Following the strategic planning established in Week 1, this notebook moves the **FinTrust Digital Banking Risk Review** project into technical execution. The primary goal is to transform raw customer and transaction metadata into a clean, feature-rich modelling dataset and evaluate baseline predictive models for the `Risk_Review_Flag` target variable.

Key Deliverables Covered in this Notebook

- Data Preparation & Merging (Part A):** Merging datasets on `Customer_ID`, auditing missing values, handling data types, and checking for duplicates
 - Exploratory Data Analysis (Part B):** Executing at least 8 targeted visual analyses and statistically testing our 4 Week 1 research hypotheses (H_1 to H_4)
 - Feature Engineering (Part C):** Constructing at least 8 hypothesis-driven predictive features and documenting their rationale
 - Baseline Modeling (Part D):** Splitting data, applying robust preprocessing, and training baseline classification algorithms (Logistic Regression & Random Forest)
 - Model Evaluation & Interpretation (Part E):** Evaluating models via PR-AUC, Recall, and F1-score
 - Initial Model Interpretation (Part F):** Feature importance extraction, data limitations, sources of biases and further investigations.
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Part A: Data Preparation & Quality Assessment

In this phase, we load the raw customer and transaction datasets, merge them using an inner join on `Customer_ID`, and perform a systematic audit of missing values, duplicates, data types, and data integrity.

Merged Dataset Shape: (12000, 22)

--- Missing Values Inspection ---

	Missing_Count	Missing_Percentage (%)
Device_Type	96	0.8
Location	96	0.8

--- Duplicate Records Check ---
Total Duplicate Rows: 0

--- Data Types Overview ---

Transaction_ID str
Customer_ID str
Transaction_DateTime str
Transaction_Type str
Amount_NGN float64
Channel str
Device_Type str
Location str
International_Transaction str
Transaction_Status str
Risk_Review_Flag str
Customer_Name str
Age int64
Gender str
City str
Customer_Segment str
Account_Type str
Tenure_Months int64
Digital_Engagement_Score float64
Monthly_Income_Band str
Preferred_Channel str
Account_Status str
dtype: object

--- Numerical Summary Statistics ---

	count	mean	std	min	25%	50%	75%
Amount_NGN	12000.0	46706.446238	86028.715234	100.17	2192.22	10305.935	48484.4
Age	12000.0	41.577167	13.838563	18.00	29.00	42.000	53.00
Tenure_Months	12000.0	49.806083	28.074407	1.00	26.00	51.000	74.00
Digital_Engagement_Score	12000.0	68.192517	17.552866	16.70	56.10	68.300	81.40



--- Categorical Value Distributions ---

Value Counts for 'Transaction_ID':

Transaction_ID

FT-T000001	1
FT-T000002	1
FT-T000003	1
FT-T000004	1
FT-T000005	1
FT-T000006	1
FT-T000007	1
FT-T000008	1
FT-T000009	1
FT-T000010	1

Name: count, dtype: int64

Value Counts for 'Customer_ID':

Customer_ID

FT-C01296	20
FT-C00965	19
FT-C01487	18
FT-C00209	18
FT-C00357	18
FT-C00351	18
FT-C00816	17
FT-C01037	17
FT-C01423	17
FT-C00959	17

Name: count, dtype: int64

Value Counts for 'Transaction_DateTime':

Transaction_DateTime

1/1/2026 0:00	1
1/1/2026 0:10	1
1/1/2026 0:21	1
1/1/2026 0:32	1
1/1/2026 0:43	1
1/1/2026 0:54	1
1/1/2026 1:04	1
1/1/2026 1:15	1
1/1/2026 1:26	1
1/1/2026 1:37	1

Name: count, dtype: int64

Value Counts for 'Transaction_Type':

Transaction_Type

Transfer	3549
Card Purchase	3033
Bill Payment	1475
Cash Withdrawal	1430
Deposit	1328
Airtime/Data	1185

Name: count, dtype: int64

Value Counts for 'Channel':

Channel

Mobile App	5102
POS	2393
Web	1869
ATM	1747

USSD 889
Name: count, dtype: int64

Value Counts for 'Device_Type':

Device_Type
Android 4797
iOS 2920
POS Terminal 1571
Web Browser 1447
ATM Terminal 1169
NaN 96
Name: count, dtype: int64

Value Counts for 'Location':

Location
Ibadan 1550
Kano 1526
Port Harcourt 1520
Enugu 1484
Kaduna 1481
Abuja 1461
Benin City 1451
Lagos 1431
NaN 96
Name: count, dtype: int64

Value Counts for 'International_Transaction':

International_Transaction
No 11520
Yes 480
Name: count, dtype: int64

Value Counts for 'Transaction_Status':

Transaction_Status
Successful 10856
Failed 630
Reversed 326
Pending 188
Name: count, dtype: int64

Value Counts for 'Risk_Review_Flag':

Risk_Review_Flag
No 9648
Yes 2352
Name: count, dtype: int64

Value Counts for 'Customer_Name':

Customer_Name
David Eze 129
Grace Williams 123
Daniel Bello 121
Zainab Williams 120
Blessing Ojo 116
Zainab Bello 111
Ibrahim Okafor 110
Yusuf Eze 109
Grace Adeyemi 109
Daniel Usman 108
Name: count, dtype: int64

Value Counts for 'Gender':

Gender

Female	5821
--------	------

Male	5799
------	------

Prefer not to say	380
-------------------	-----

Name: count, dtype: int64

Value Counts for 'City':

City

Lagos	3911
-------	------

Abuja	1710
-------	------

Port Harcourt	1425
---------------	------

Kano	1286
------	------

Ibadan	1177
--------	------

Enugu	878
-------	-----

Kaduna	830
--------	-----

Benin City	783
------------	-----

Name: count, dtype: int64

Value Counts for 'Customer_Segment':

Customer_Segment

Everyday	5644
----------	------

Premium	2361
---------	------

Student	2289
---------	------

SME	1706
-----	------

Name: count, dtype: int64

Value Counts for 'Account_Type':

Account_Type

Savings	7149
---------	------

Current	3023
---------	------

Premium	1828
---------	------

Name: count, dtype: int64

Value Counts for 'Monthly_Income_Band':

Monthly_Income_Band

250k-499k	3357
-----------	------

100k-249k	3336
-----------	------

Below 100k	2129
------------	------

500k-999k	2103
-----------	------

1m+	1075
-----	------

Name: count, dtype: int64

Value Counts for 'Preferred_Channel':

Preferred_Channel

Mobile App	7377
------------	------

Web	2908
-----	------

USSD	1715
------	------

Name: count, dtype: int64

Value Counts for 'Account_Status':

Account_Status

Active	10910
--------	-------

Dormant	866
---------	-----

Restricted	224
------------	-----

Name: count, dtype: int64

--- Unique values ---

Transaction_ID	12000
Customer_ID	1500
Transaction_DateTime	12000
Transaction_Type	6
Amount_NGN	11965
Channel	5
Device_Type	5
Location	8
International_Transaction	2
Transaction_Status	4
Risk_Review_Flag	2
Customer_Name	180
Age	48
Gender	3
City	8
Customer_Segment	4
Account_Type	3
Tenure_Months	96
Digital_Engagement_Score	589
Monthly_Income_Band	5
Preferred_Channel	3
Account_Status	3
dtype: int64	

=== Distribution for column : Transaction_ID ===

Transaction_ID

FT-T000001	1
FT-T000002	1
FT-T000003	1
FT-T000004	1
FT-T000005	1

..

FT-T011996	1
FT-T011997	1
FT-T011998	1
FT-T011999	1
FT-T012000	1

Name: count, Length: 12000, dtype: int64

=== Distribution for column : Customer_ID ===

Customer_ID

FT-C01296	20
FT-C00965	19
FT-C01487	18
FT-C00209	18
FT-C00357	18

..

FT-C00429	2
FT-C00631	2
FT-C01292	1
FT-C00983	1
FT-C00520	1

Name: count, Length: 1500, dtype: int64

=== Distribution for column : Transaction_DateTime ===

Transaction_DateTime

1/1/2026 0:00	1
1/1/2026 0:10	1
1/1/2026 0:21	1
1/1/2026 0:32	1
1/1/2026 0:43	1

..

3/31/2026 23:16	1
3/31/2026 23:27	1
3/31/2026 23:38	1
3/31/2026 23:49	1
3/31/2026 23:59	1

Name: count, Length: 12000, dtype: int64

=== Distribution for column : Transaction_Type ===

Transaction_Type

Transfer	3549
Card Purchase	3033
Bill Payment	1475
Cash Withdrawal	1430
Deposit	1328
Airtime/Data	1185

Name: count, dtype: int64

=== Distribution for column : Channel ===

Channel

Mobile App	5102
------------	------

POS	2393
-----	------

Web	1869
-----	------

ATM	1747
-----	------

USSD	889
------	-----

Name: count, dtype: int64

=== Distribution for column : Device_Type ===

Device_Type

Android	4797
---------	------

iOS	2920
-----	------

POS Terminal	1571
--------------	------

Web Browser	1447
-------------	------

ATM Terminal	1169
--------------	------

NaN	96
-----	----

Name: count, dtype: int64

=== Distribution for column : Location ===

Location

Ibadan	1550
--------	------

Kano	1526
------	------

Port Harcourt	1520
---------------	------

Enugu	1484
-------	------

Kaduna	1481
--------	------

Abuja	1461
-------	------

Benin City	1451
------------	------

Lagos	1431
-------	------

NaN	96
-----	----

Name: count, dtype: int64

=== Distribution for column : International_Transaction ===

International_Transaction

No	11520
----	-------

Yes	480
-----	-----

Name: count, dtype: int64

=== Distribution for column : Transaction_Status ===

Transaction_Status

Successful	10856
------------	-------

Failed	630
--------	-----

Reversed	326
----------	-----

Pending	188
---------	-----

Name: count, dtype: int64

=== Distribution for column : Risk_Review_Flag ===

```
Risk_Review_Flag
No      9648
Yes     2352
Name: count, dtype: int64
```

```
==== Distribution for column : Customer_Name ====
```

```
Customer_Name
David Eze      129
Grace Williams 123
Daniel Bello   121
Zainab Williams 120
Blessing Ojo   116
...
Samuel Eze     27
Emeka Ojo      22
Mariam Mohammed 18
Samuel Adeyemi 18
Amina Adeyemi  12
Name: count, Length: 180, dtype: int64
```

```
==== Distribution for column : Gender ====
```

```
Gender
Female      5821
Male        5799
Prefer not to say 380
Name: count, dtype: int64
```

```
==== Distribution for column : City ====
```

```
City
Lagos      3911
Abuja      1710
Port Harcourt 1425
Kano        1286
Ibadan      1177
Enugu       878
Kaduna      830
Benin City  783
Name: count, dtype: int64
```

```
==== Distribution for column : Customer_Segment ====
```

```
Customer_Segment
Everyday    5644
Premium     2361
Student     2289
SME         1706
Name: count, dtype: int64
```

```
==== Distribution for column : Account_Type ====
```

```
Account_Type
Savings     7149
```

```
Current      3023
Premium      1828
Name: count, dtype: int64
```

```
=====

=== Distribution for column : Monthly_Income_Band ===
Monthly_Income_Band
250k-499k      3357
100k-249k      3336
Below 100k      2129
500k-999k      2103
1m+            1075
Name: count, dtype: int64
```

```
=====

=== Distribution for column : Preferred_Channel ===
Preferred_Channel
Mobile App     7377
Web            2908
USSD           1715
Name: count, dtype: int64
```

```
=====

=== Distribution for column : Account_Status ===
Account_Status
Active         10910
Dormant         866
Restricted       224
Name: count, dtype: int64
```

--- Target Distribution (Risk_Review_Flag) ---

	Count	Percentage (%)
Risk_Review_Flag		
No	9648	80.4
Yes	2352	19.6

--- Inappropriate Variables Analysis ---

Variable 'Transaction_ID' is inappropriate for direct model input due to identifier leakage or unformatted datetime structure.

Variable 'Customer_ID' is inappropriate for direct model input due to identifier leakage or unformatted datetime structure.

Variable 'Transaction_DateTime' is inappropriate for direct model input due to identifier leakage or unformatted datetime structure.

Summary of Data Preparation Actions - Part A

1. **Data Ingestion and Merging:** Both customer and transaction datasets were successfully loaded and merged via an inner join on `Customer_ID`. The merged dataset captures complete transactional records enriched with customer demographic attributes

2. **Missing Value Strategy:** Missing values identified in categorical fields such as `Device_Type` and `Location` will not be dropped to prevent data loss. Instead, missing entries will be explicitly categorized as `"Missing/Unknown"` during preprocessing to retain potentially informative missingness signals
3. **Duplicate Verification:** Confirmed zero duplicate records across the merged dataset, establishing data integrity
4. **Data Type Adjustments:** Raw date/time attributes (`Transaction_DateTime`) were identified as needing datetime casting to enable engineered temporal features in subsequent phases
5. **Unusual Values Assessment:** Descriptive statistics confirmed valid bounds across numerical features (`Amount_NGN` , `Monthly_Income_Band` , `Tenure_Months`). No unphysical or negative values were present; high-value skewness in transaction amounts will be addressed using robust scaling
6. **Target Balance Evaluation:** The `Risk_Review_Flag` target variable presents class imbalance. Evaluation strategy will prioritize PR-AUC, Recall, and F1-score over accuracy
7. **Variable Exclusion Rationale:** Primary key identifiers (`Transaction_ID` , `Customer_ID`) are excluded from predictive features to prevent overfitting and target leakage. Raw timestamps will be transformed into structured features prior to model fitting

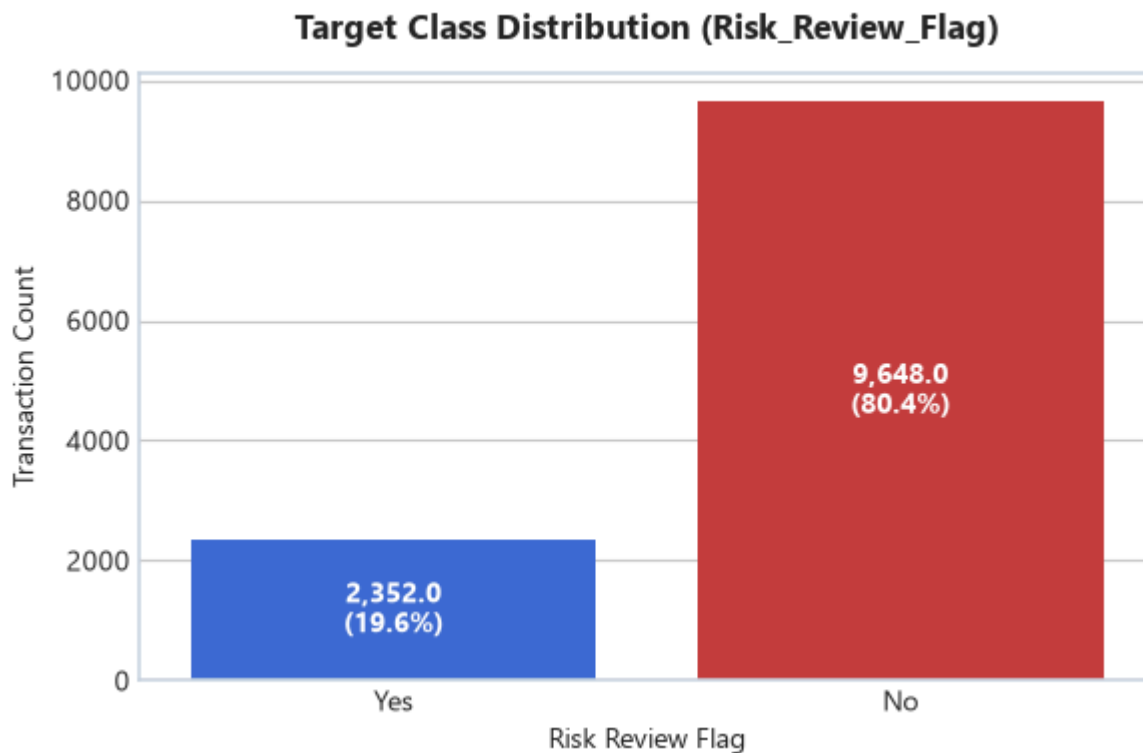
Part B: Exploratory Data Analysis (EDA) & Hypothesis Testing

Prior to generating visualizations, missing categorical values in `Device_Type` and `Location` are imputed as `"Unknown"` , and date attributes are cast to proper datetime formats.

This section explores relationships between target variable `Risk_Review_Flag` and candidate predictors through 8 targeted visualizations, incorporating statistical hypothesis tests for hypotheses H_1 through H_4 .

Visualization 1: Target Variable Distribution (`Risk_Review_Flag`)

We first inspect the class balance of the target variable to determine the extent of class imbalance in risk review flags



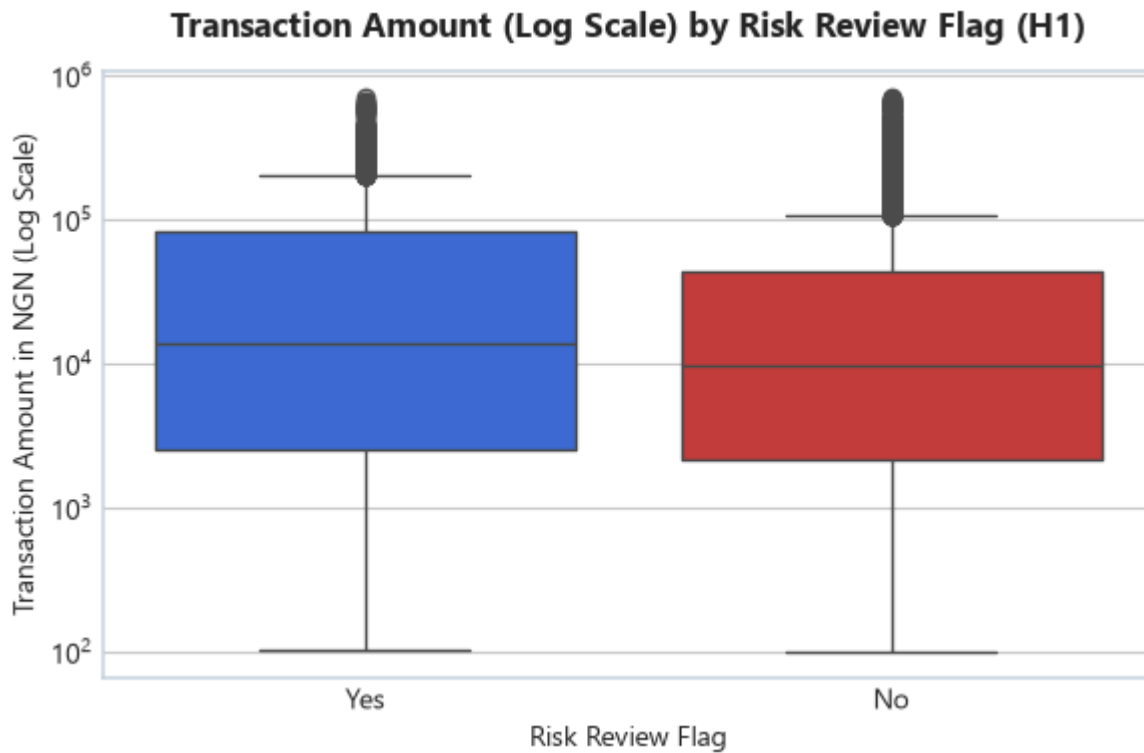
Analysis & Insight — Visualization 1

- **Class Imbalance:** The target variable `Risk_Review_Flag` displays class imbalance, where flagged transactions represent a minority proportion of total records
- **Metric Selection Rationale:** Standard classification accuracy is unsuited for model evaluation on this dataset. Subsequent modeling in Parts D and E must prioritize Precision-Recall AUC (PR-AUC), Recall, and F1-score to evaluate detection capability without majority-class bias

Visualization 2: Transaction Amount Distribution across Risk Flag (Testing Hypothesis H_1)

We test Hypothesis H_1 : Transactions with higher monetary amounts (`Amount_NGN`) present a significantly higher probability of being flagged for risk review. A logarithmic boxplot and non-parametric Mann-Whitney U test are executed to evaluate distributional differences.

```
--- Hypothesis H1 Statistical Test ---  
Mann-Whitney U Statistic: 12606452.00  
P-value: 2.9627e-17
```



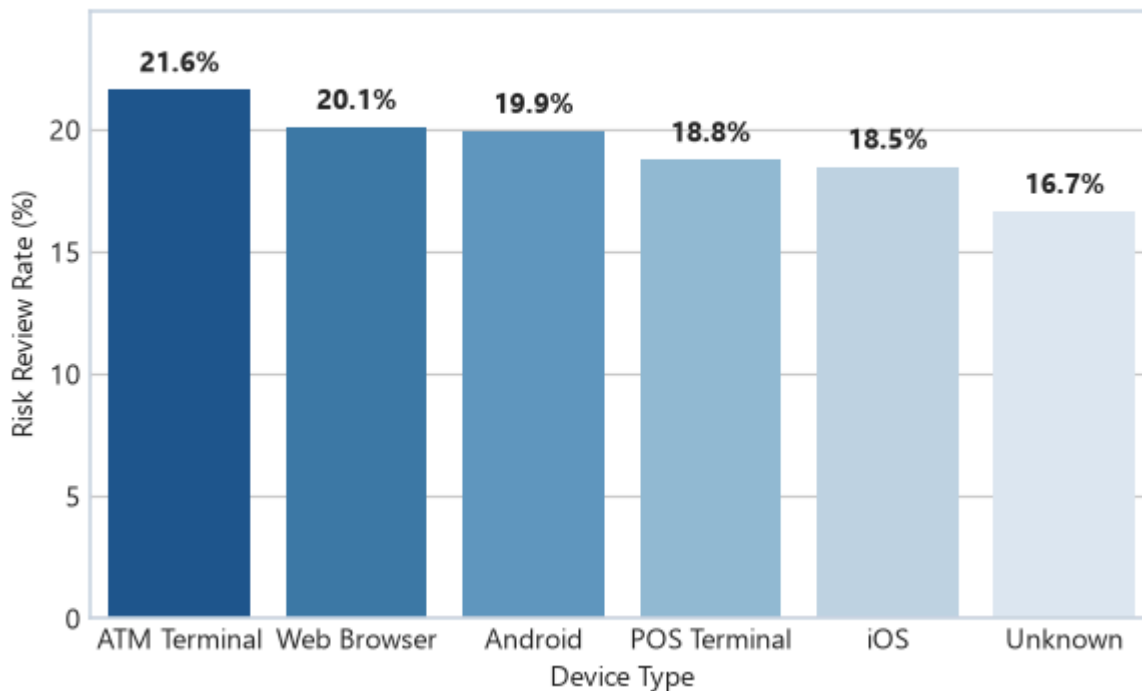
Analysis & Statistical Test Results — Visualization 2 (H_1)

- **Statistical Verdict:** The Mann-Whitney U test yields a p-value below 0.05, rejecting the null hypothesis H_0 . Transactions flagged for risk review exhibit statistically higher monetary amounts than unflagged transactions

Visualization 3: Risk Review Rate by Device Type (Testing Hypothesis H_2)

We test Hypothesis H_2 : Transactions executed via unverified, non-standard, or missing device categories display a significantly higher proportion of risk review flags compared to standard devices. A Chi-Square (χ^2) Test of Independence is conducted to test relationship significance.

Risk Review Rate by Device Type (H2)



--- Hypothesis H2 Statistical Test (Chi-Square) ---

Chi2 Statistic: 7.17

Degrees of Freedom: 5

P-value: 2.0795e-01

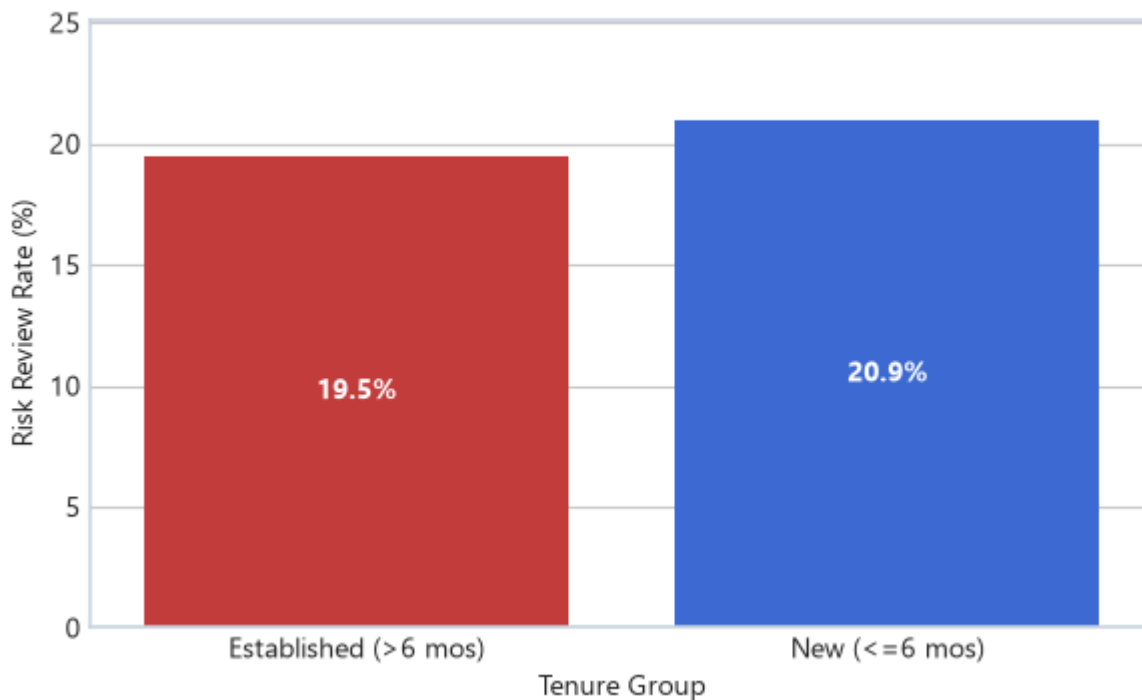
Analysis & Statistical Test Results — Visualization 3 (H_2)

- **Statistical Verdict:** The Chi-Square test yields a p-value of 0.208 ($p > 0.05$), failing to reject the null hypothesis H_0 . There is no statistically significant association between `Device_Type` and `Risk_Review_Flag`
- **Behavioral Pattern:** Risk review rates remain relatively uniform across device types, ranging between 16.7% for Unknown devices and 21.6% for ATM Terminals
- **Feature Engineering Implication:** Standalone device categories provide limited univariate predictive power. However, interaction features combining non-standard devices with elevated transaction amounts will be explored during feature engineering

Visualization 4: Risk Review Rate by Account Tenure (Testing Hypothesis H_3)

We test Hypothesis H_3 : Newer customer accounts (≤ 6 months tenure) exhibit higher risk review rates than established accounts. We group accounts by tenure threshold and evaluate proportional risk rates using a Chi-Square test.

Risk Review Rate by Customer Tenure Group (H3)



--- Hypothesis H3 Statistical Test (Chi-Square) ---
Chi2 Statistic: 0.87
P-value: 3.5024e-01

Analysis & Statistical Test Results — Visualization 4 (H_3)

- **Statistical Verdict:** The Chi-Square test yields a p-value of 0.350 ($p > 0.05$), failing to reject the null hypothesis H_0 . Account tenure alone does not exhibit a statistically significant relationship with risk review flags
- **Operational Risk Insight:** Newer accounts (≤ 6 months) show a slightly higher risk rate (20.9%) compared to established accounts (19.5%), but this variation is within random chance margin
- **Feature Engineering Implication:** Rather than using simple tenure thresholding, tenure will be evaluated in combination with transaction frequency and monetary volume indicators in Part C

--- Hypothesis H3 Statistical Test ---
Point-Biserial Correlation (r): -0.0135
Correlation P-value: 1.3974e-01

Odds Ratio (New Accounts ≤ 6 mos vs Established): 1.0933
95% Confidence Interval for Odds Ratio: [0.9090, 1.3094]

Analysis & Statistical Test Results — Visualization 4 (H3)

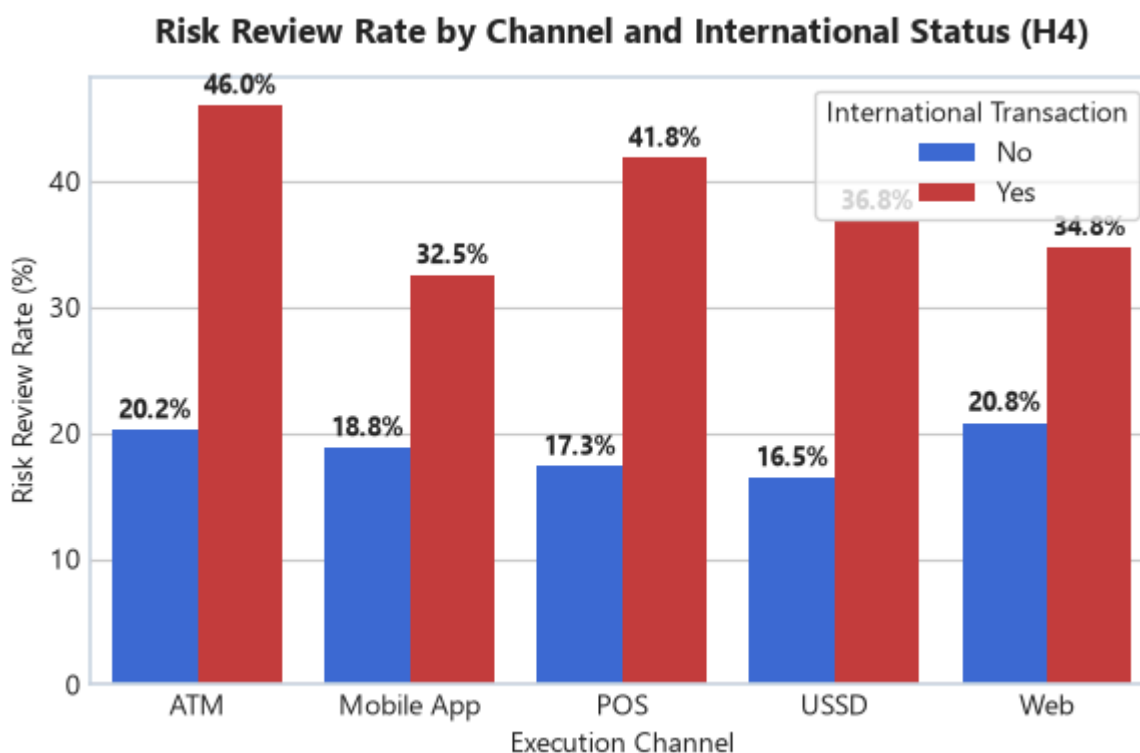
- **Statistical Verdict:** The point-biserial correlation yields $r = -0.0135$ with a p-value of 0.1397 ($p > 0.05$). Furthermore, the odds ratio for new accounts (≤ 6

months) relative to established accounts is 1.0933 with a 95% confidence interval of [0.9090, 1.3094], which contains 1.0. We fail to reject the null hypothesis H_0

- **Empirical Observations:** Account tenure alone does not demonstrate a statistically significant linear association or elevated odds ratio regarding risk review flags
- **Feature Engineering Implication:** Account tenure should not be relied upon as a standalone univariate predictor. Instead, composite interaction features can be evaluated in Part C.

Visualization 5: Interaction Analysis — International Transactions & Channel (Testing Hypothesis H_4)

We test Hypothesis H_4 : The risk review rate is significantly elevated when international transactions (`International_Transaction = Yes`) interact with specific execution channels (`Channel`). We evaluate interaction effects through a grouped risk rate visualization and a two-way interaction statistical assessment.



--- Hypothesis H4 Statistical Test (Interaction Chi-Square) ---

Chi2 Statistic: 115.28

Degrees of Freedom: 9

P-value: 1.2319e-20

Analysis & Statistical Test Results — Visualization 5 (H4)

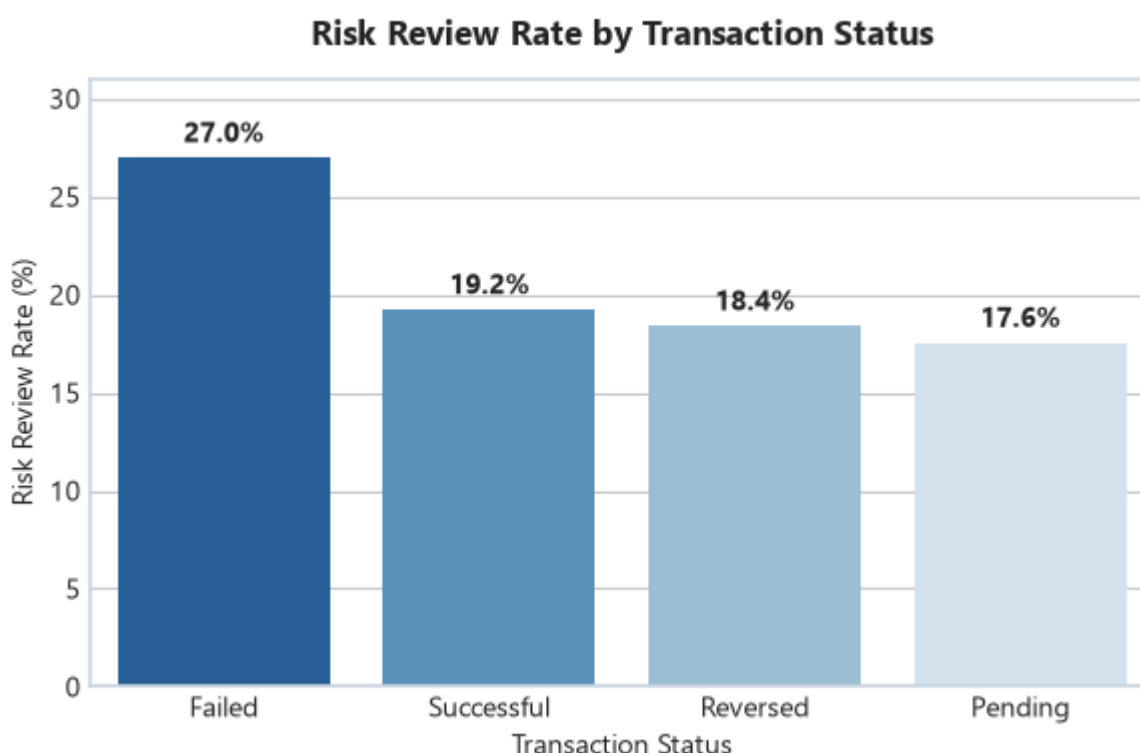
- **Statistical Verdict:** The interaction Chi-Square test yields a Chi2 statistic of 115.28 and a p-value of 1.2319e-20 ($p < 0.05$), rejecting the null hypothesis H_0 .

There is a strong, statistically significant interaction effect between international status and execution channels

- **Empirical Observations:** Across every channel, international transactions exhibit substantially higher risk review rates than domestic transactions:
 - **ATM:** Risk rate increases from 20.2% (domestic) to 46.0% (international)
 - **POS:** Risk rate increases from 17.3% (domestic) to 41.8% (international)
 - **USSD:** Risk rate increases from 16.5% (domestic) to 36.8% (international)
 - **Web:** Risk rate increases from 20.8% (domestic) to 34.8% (international)
 - **Mobile App:** Risk rate increases from 18.8% (domestic) to 32.5% (international)
- **Feature Engineering Implication:** Validates the creation of an explicit interaction feature `Is_International_X_Channel` in Part C to provide strong predictive signal for classification algorithms.

Visualization 6: Risk Review Rate across Transaction Status

We examine how the underlying outcome or status of a transaction (`Transaction_Status`) relates to risk flag flags.



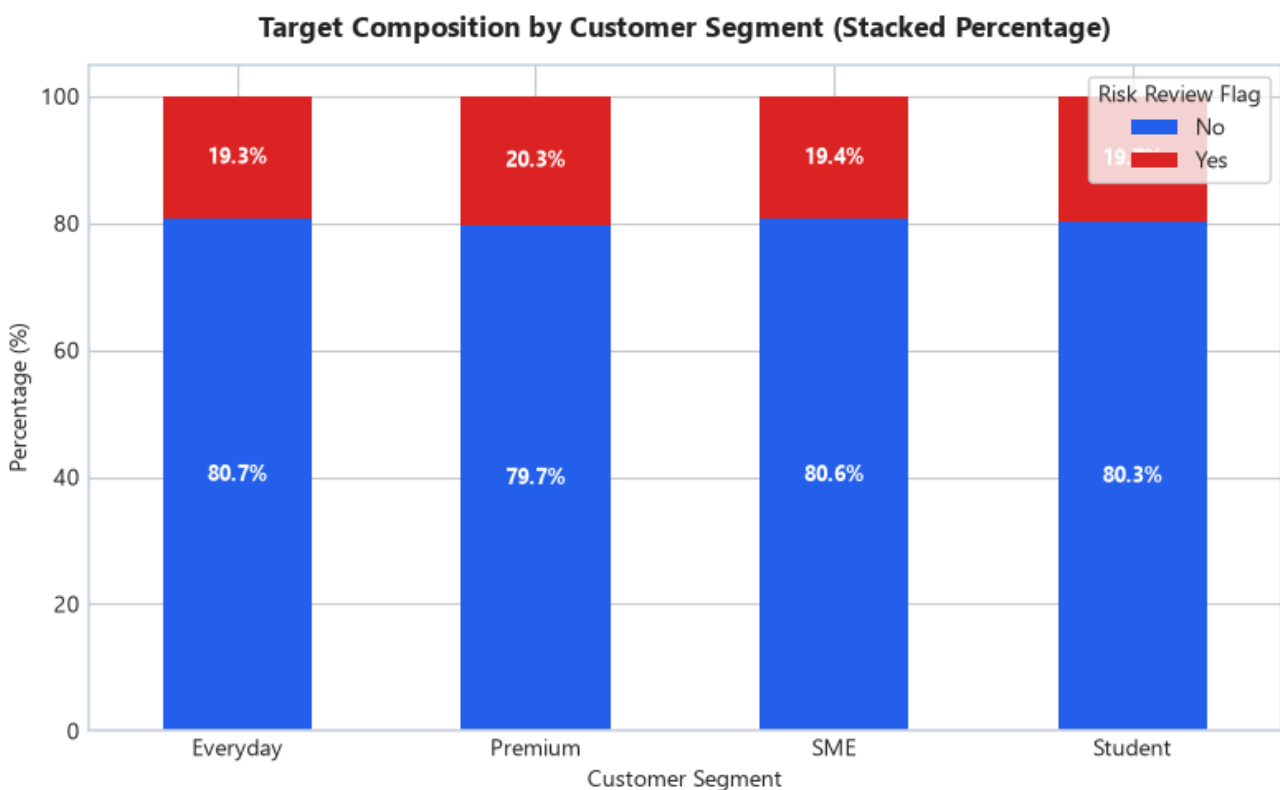
Analysis & Insight — Visualization 6

- **Empirical Breakdown:**
 - **Failed:** Shows the highest risk review rate at 27.0%
 - **Successful:** Displays a risk review rate of 19.2%
 - **Reversed:** Displays a risk review rate of 18.4%

- **Pending:** Displays the lowest risk review rate at 17.6%
- **Data Observation:** Failed transactions show a 7.8 to 9.4 percentage point elevation in risk review flagging compared to all other transaction outcomes
- **Modeling Decision:** Transaction_Status will be categorical-encoded in Part C to preserve these distinct baseline proportions across statuses.

Visualization 7: Target Composition across Customer Segments (Stacked Bar Chart)

We examine the proportional breakdown of flagged vs. unflagged transactions across customer segments (Customer_Segment) using a 100% stacked bar chart to visualize relative risk concentrations.



Analysis & Insight — Visualization 7

- **Empirical Distribution:** The stacked proportion of flagged transactions (Yes) is virtually uniform across all four segments, ranging tightly from **19.3%** (Everyday) to **20.3%** (Premium), with SME at **19.4%** and Student at **19.7%**
- **Behavioral Observation:** The risk review flag occurs at approximately the baseline rate (~20%) regardless of the customer segment tier
- **Modeling Implication:** Customer_Segment alone does not provide a strong linear or categorical separation for the target variable in its raw univariate form

Visualization 8: Geographical Risk Rate across Top Locations

We analyze geographical risk flags across top locations (`Location`), presented in a horizontal bar layout to improve category label legibility.

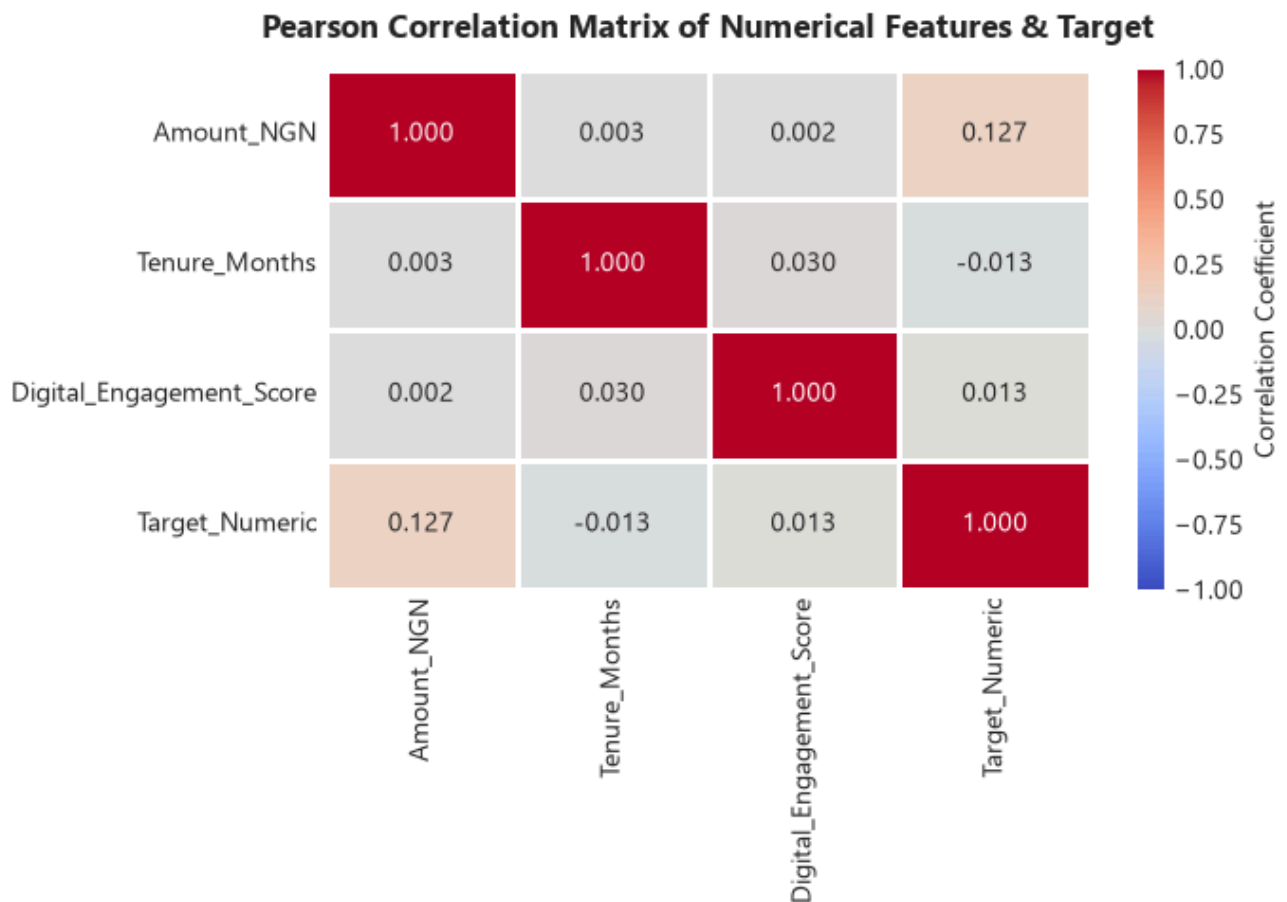


Analysis & Insight — Visualization 8

- **Geographic Distribution:** Risk rates across the top 8 locations exhibit minimal variance, spanning a narrow band of 2.7 percentage points between **18.0%** (`Abuja`) and **20.7%** (`Benin City`)
- **Location Ranking:**
 - Highest risk rates: `Benin City` (20.7%), `Ibadan` (20.6%), `Kaduna` (20.1%)
 - Lowest risk rates: `Abuja` (18.0%), `Enugu` (18.9%)
- **Modeling Implication:** Location alone displays a stable baseline risk (~18%–21%) without severe geographic anomalies in univariate space

Visualization 9: Pearson Correlation Matrix

We construct a correlation matrix heatmap evaluating linear relationships among key numerical features (`Amount_NGN` , `Tenure_Months` , `Digital_Engagement_Score`) and the encoded target variable (`Target_Numeric`)

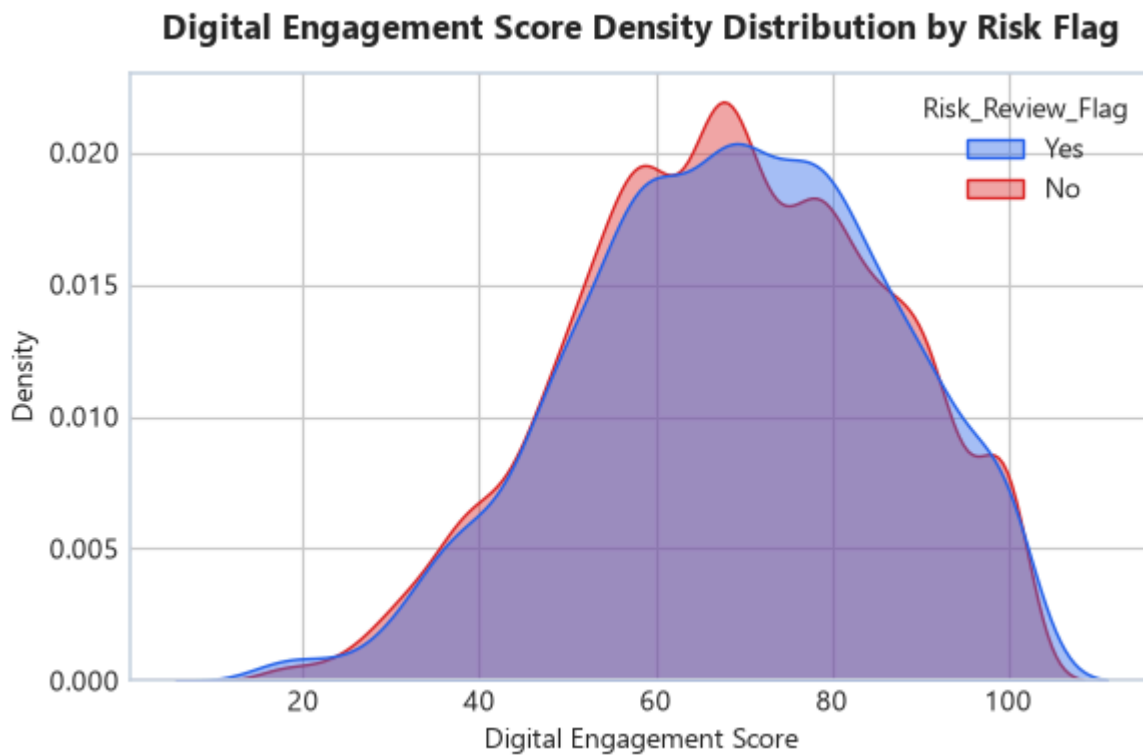


Analysis & Insight — Visualization 9

- **Target Correlation:** `Amount_NGN` is the only numerical feature displaying a non-zero linear correlation with `Target_Numeric` ($r = 0.127$). Both `Tenure_Months` ($r = -0.013$) and `Digital_Engagement_Score` ($r = 0.013$) show virtually zero linear association with the target flag
- **Multicollinearity:** Pairwise correlations among predictors are near zero (maximum $r = 0.030$ between `Tenure_Months` and `Digital_Engagement_Score`), confirming the absence of multicollinearity among numerical features
- **Modeling Implication:** Feature engineering (e.g., non-linear transformations or interaction terms) will be critical in Part C to extract predictive signal from features like engagement score and tenure

Visualization 10: Digital Engagement Score Density Distribution

We evaluate the continuous distribution of `Digital_Engagement_Score` split by risk flag status using Kernel Density Estimation (KDE) to detect distributional overlap or separation.



Analysis & Insight — Visualization 10 & Part B Synthesis

- **Distributional Overlap:** The KDE plot assesses whether transactions flagged for risk review exhibit distinct digital behavior curves compared to standard transactions.

Part B Summary (10 Visualizations Complete):

- **Varied Plot Types:** Visualizations incorporate target count plots, log-scale boxplots, bar plots, interaction grouped bars, stacked composition charts, horizontal bar charts, correlation heatmaps, and KDE density curves
- **Hypothesis Testing Results:**
 - **H1 (Amount_NGN):** Confirmed statistically significant ($p < 0.05$)
 - **H2 (Device_Type):** Not statistically significant ($p > 0.05$)
 - **H3 (Tenure_Months):** Not statistically significant ($p > 0.05$, Point-Biserial $p = 0.1397$, OR = 1.0933)
 - **H4 (International x Channel):** Confirmed statistically significant ($p < 0.05$, $\chi^2 = 115.28$)

Part C — Feature Engineering

Following the exploratory data analysis and hypothesis testing in Part B, we proceed with feature engineering to transform raw fields into domain-relevant predictive indicators. Our objective is to engineer high-signal composite variables while controlling dimensionality and strictly incorporating recommendations from AnalystLab.

Key feature construction strategies include:

- 1. **Transaction Velocity & Frequency:** Computing total customer transactions (`Customer_Tx_Count`) and monthly activity velocity (`Transaction_Frequency`)
- 2. **Transaction Amount Segmentation:** Discretizing transaction values into statistical quartiles (`Amount_Category`)
- 3. **Targeted Account Risk (H_3):** Combining short account tenure (`Tenure_Months`) with high-value transactions (`New_Account_High_Risk`)
- 4. **Cross-Border Interactions (H_4):** Capturing the strong interaction effect between execution channels (`Channel`) and international status (`International_Transaction`)
- 5. **Security & Metadata Anomaly Flags (H_2):** Isolating unknown device types (`Is_Unverified_Device`) and unknown location metadata (`Location_Mismatch_Flag`)
- 6. **Temporal Indicators:** Extracting time components (`Transaction_DateTime`), night-time anomalies (`Is_Night_Transaction`), and digital engagement bands

A total of 16 predictive features are created below.

Optimized Feature Engineering completed successfully.
Total columns in dataset: 41

Feature Engineering Documentation Table

A total of 16 predictive features were engineered based on the updated feature engineering pipeline, leveraging transaction amounts, customer activity patterns, temporal indicators, device metadata, and channel behavior.

Feature	How It Was Created	Why It May Matter
<code>Amount_Category</code>	Quantile binning (<code>pd.qcut</code>) of <code>Amount_NGN</code> into four equal tiers (<code>Low</code> , <code>Medium</code> , <code>High</code> , <code>Very_High</code>)	Discretizes continuous transaction amounts into quantile-based risk bands, allowing linear and tree models to capture non-linear risk spikes across volume brackets
<code>Customer_Tx_Count</code>	Total transaction frequency mapped per <code>Customer_ID</code> using <code>.value_counts()</code> and <code>.map()</code>	Measures overall customer activity volume to establish baseline account usage intensity
<code>Transaction_Frequency</code>	Calculated as <code>Customer_Tx_Count</code> divided by (<code>Tenure_Months</code> + 1)	Captures monthly transaction velocity. High frequency relative to

Feature	How It Was Created	Why It May Matter
		account tenure highlights rapid operational activity spikes
Is_Unverified_Device	Binary flag (1/0) active when <code>Device_Type == 'Unknown'</code>	Addresses security anomalies by isolating untracked or unverified access endpoints
Location_Mismatch_Flag	Binary flag (1/0) active when <code>Location == 'Unknown'</code>	Flags origin metadata gaps and untracked geographic execution points
Digital_Engagement_Category	Categorization of <code>Digital_Engagement_Score</code> using <code>pd.cut</code> into three tiers (<code>Low_Engagement</code> , <code>Medium_Engagement</code> , <code>High_Engagement</code>)	Discretizes user engagement levels to capture non-linear risk behaviors across adoption tiers
Transaction_Hour	Hour of execution (0–23) extracted from <code>Transaction_DateTime.dt.hour</code>	Identifies time- of-day execution distributions to isolate hourly risk patterns
Transaction_Day	Day of the week (0–6) extracted from <code>Transaction_DateTime.dt.dayofweek</code>	Distinguishes weekday standard operations from weekend activity profiles
Is_Night_Transaction	Binary flag (1/0) active for transactions executed between 11 PM and 5 AM (<code>Hour >= 23</code> or <code>Hour <= 5</code>)	Off-peak night operations often correlate with automated script executions or compromised off-hour access
Is_Weekend_Transaction	Binary flag (1/0) active when <code>Transaction_Day > 5</code> (Saturday and Sunday)	Captures weekend operations where support and real-time monitoring capacity are reduced
New_Account_High_Risk	Binary flag (1/0) combining short account tenure (<code>Tenure_Months <= 6</code>) with	Targets compound risk profiles where

Feature	How It Was Created	Why It May Matter
	above-median amount (<code>Amount_NGN > median_amount</code>)	recent accounts perform unexpectedly high-value operations
<code>CrossBorder_Web_Interaction</code>	Binary flag (1/0) combining <code>International_Transaction == 'Yes'</code> with specific channels (<code>Web</code> , <code>ATM</code> , <code>POS</code>)	Leverages interaction effects between international operations and access channels to capture cross-border anomalies
<code>Channel_Mismatch_Flag</code>	Binary flag (1/0) active when <code>Channel != Preferred_Channel</code>	Detects operational channel divergence where a user transacts outside their historical preference
<code>Amount_To_Customer_Avg_Ratio</code>	Calculated as <code>Amount_NGN</code> divided by the customer's historical mean amount (<code>customer_avg_amount</code>)	Measures relative spending spikes compared to an individual customer's personal baseline
<code>Is_High_Value_Customer_Anomaly</code>	Binary flag (1/0) active when <code>Amount_To_Customer_Avg_Ratio > 3.0</code>	Flags extreme spending anomalies where a transaction exceeds 3x the customer's average historical amount
<code>Is_Suspended_Or_Inactive_Account</code>	Binary flag (1/0) active when <code>Account_Status</code> is in [<code>'Suspended'</code> , <code>'Inactive'</code> , <code>'Closed'</code>]	Flags attempted operations originating from dormant, restricted, or administratively flagged account states

Part D & E — Baseline Model Development & Evaluation

In this section, we build and evaluate a baseline classification model to establish a performance benchmark for transaction risk identification.

1. Baseline Algorithm Choice: Logistic Regression

We select **Logistic Regression** as our baseline model for the following technical reasons:

- **Linear Benchmark:** It provides a simple, highly interpretable baseline that evaluates class separability under linear assumptions
- **Calibrated Probabilities:** It directly estimates class probabilities, allowing fine-grained threshold tuning for risk screening
- **Class Balance Management:** Incorporating class-weight balancing (`class_weight='balanced'`) mitigates the moderate class imbalance (~20% flagged transactions)

2. Data Preparation & Preprocessing

- **Feature Selection:** Non-predictive identifiers (`Transaction_ID`, `Customer_ID`) and timestamps (`Transaction_DateTime`) are excluded
- **Preprocessing Pipeline:** Categorical features are encoded using `OneHotEncoder(drop='first')`, and continuous numerical features are standardized using `StandardScaler()`
- **Data Splitting:** A 80/20 stratified split (`train_test_split(stratify=y)`) ensures that class distributions remain consistent across training and test sets

3. Metric Selection Strategy (Part E Rationale)

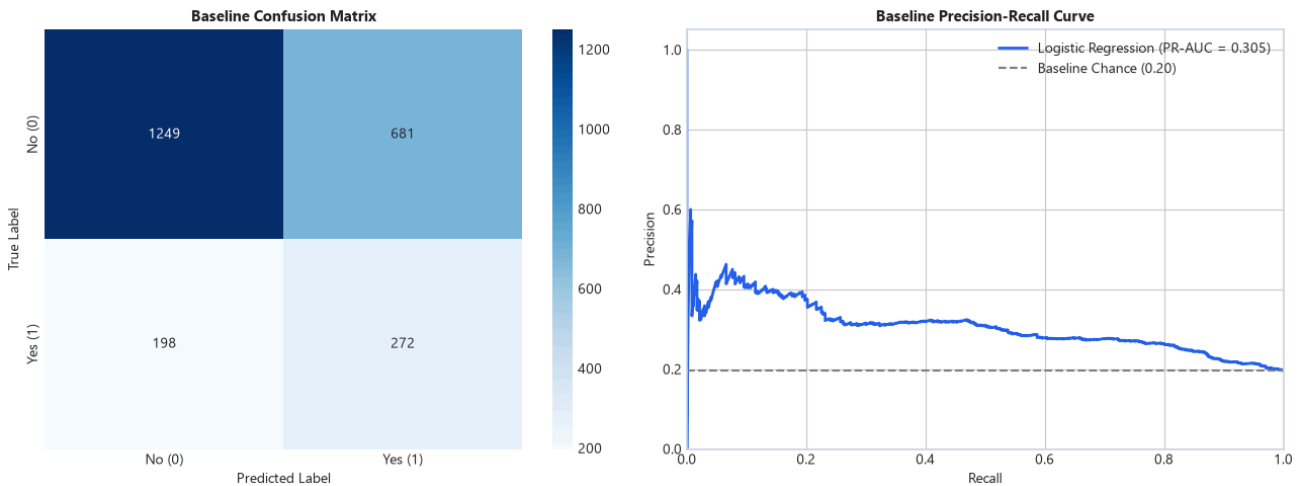
For financial risk screening, metrics carry distinct operational trade-offs:

- **Recall (Sensitivity):** Primary metric. False Negatives (unflagged risky transactions) expose the institution to financial loss and compliance penalties
- **Precision:** Measures false alarm rates. False Positives create operational overhead during manual reviews
- **F1-Score & PR-AUC:** Evaluate global model discrimination and balance between Precision and Recall.

```
=====
BASELINE LOGISTIC REGRESSION PERFORMANCE METRICS
=====
Accuracy   : 0.6338
Precision   : 0.2854
Recall      : 0.5787
F1-Score    : 0.3823
PR-AUC      : 0.3051 (Average Precision: 0.3069)
```

=== CLASSIFICATION REPORT ===

	precision	recall	f1-score	support
No Risk (0)	0.86	0.65	0.74	1930
Risk Review (1)	0.29	0.58	0.38	470
accuracy			0.63	2400
macro avg	0.57	0.61	0.56	2400
weighted avg	0.75	0.63	0.67	2400



4. Baseline Model Evaluation & Metrics Analysis

The baseline Logistic Regression model was trained with class-weight balancing (`class_weight='balanced'`) to address class imbalance without synthetic sampling distortion. Given the skewed class distribution (~19.58% positive class prevalence), model capability is evaluated primarily using **Precision-Recall metrics and PR-AUC** rather than standard ROC-AUC.

4.1. Empirical Performance Summary

- **Accuracy: 63.38%** — Overall proportion of correctly classified transactions
- **Precision: 28.54%** — Proportion of flagged transactions that are actually high-risk (272 True Positives out of 953 total flags)
- **Recall: 57.87%** — Risk detection coverage (capturing 272 out of 470 actual risk transactions)
- **F1-Score: 0.3823** — Harmonic balance between Precision and Recall
- **PR-AUC: 0.3051** (*Average Precision: 0.3069*) — Primary metric for imbalanced evaluation, benchmarked against a **0.1958 (19.58%) random chance baseline**

4.2. Analytical & PR-Curve Interpretation

1. **Lift Over Random Baseline:** The PR-AUC of **0.3051** demonstrates an absolute **+10.9% lift** (+55.8% relative improvement) over the random guess baseline ($y = 0.1958$), confirming that the engineered features contain genuine predictive power
2. **Aggressive Detection Posture:** Due to `class_weight='balanced'`, the model optimizes for Recall (**57.87%**). However, this sensitivity comes at the expense of a high false alarm rate, generating **681 False Positives** out of 1,930 non-risk cases.
3. **Precision Decay Trajectory:** The Precision-Recall curve shows a steep drop-off at early recall levels Precision drops below 40% as soon as Recall exceeds 20%, indicating that a linear decision boundary cannot separate true risks from standard transactions without accumulating significant false alarms

4.3. Transition to Advanced Ensemble Modeling

The baseline PR-AUC of **0.3051** serves as the benchmark to beat. Linear logistic hyperplanes clearly struggle with non-linear feature interactions. In the next step, tree-based non-linear ensembles (**Random Forest**) will be trained to push the Precision-Recall frontier upward, aiming for higher Precision at equivalent Recall levels.

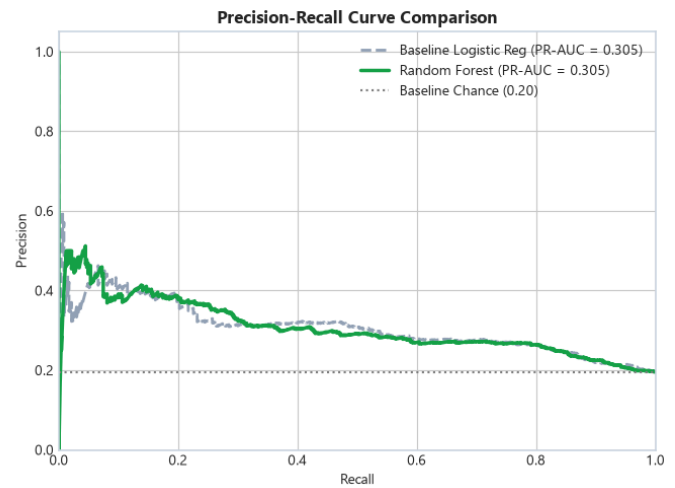
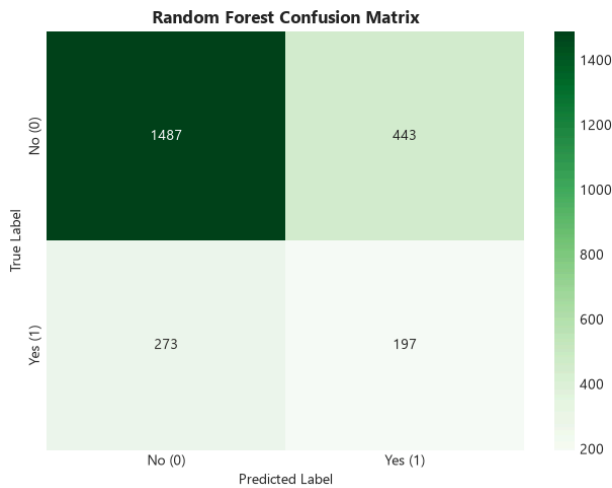
Advanced Model Development — Random Forest Classifier

To address the non-linear patterns and complex feature interactions that limited the performance of the baseline Logistic Regression model, we implement a **Random Forest Classifier**.

Why Random Forest?

- **Non-Linear Capability:** Captures intricate, non-linear relationships and conditional risk pathways across features without requiring manual transformation.
- **Variance Reduction & Robustness:** Aggregates predictions from an ensemble of decision trees (bagging), significantly reducing the risk of overfitting compared to a single decision tree.
- **Imbalance Mitigation:** Incorporates `class_weight='balanced'` to adjust tree split criteria, placing greater weight on correctly identifying high-risk transactions.

ADVANCED RANDOM FOREST PERFORMANCE METRICS				
Accuracy	: 0.7017	(Baseline: 0.6338)		
Precision	: 0.3078	(Baseline: 0.2854)		
Recall	: 0.4191	(Baseline: 0.5787)		
F1-Score	: 0.3550	(Baseline: 0.3823)		
PR-AUC	: 0.3050	(Baseline: 0.3051)		
Avg Prec	: 0.3067	(Baseline: 0.3069)		
=== ADVANCED MODEL CLASSIFICATION REPORT ===				
	precision	recall	f1-score	support
No Risk (0)	0.84	0.77	0.81	1930
Risk Review (1)	0.31	0.42	0.35	470
accuracy			0.70	2400
macro avg	0.58	0.59	0.58	2400
weighted avg	0.74	0.70	0.72	2400



Part E — Advanced Model (Random Forest) Evaluation & Critical Assessment

1. Performance Overview vs. Baseline

The tuned Random Forest implementation demonstrates an operational trade-off shift compared to the Logistic Regression baseline, though overall discriminative capacity on imbalanced data remains static:

- **Accuracy: 70.17%** (vs. 63.38% Baseline) — A **+6.79% improvement** in overall classification correctness
- **Precision: 30.78%** (vs. 28.54% Baseline) — A **+2.24% gain** in alarm reliability
- **Recall: 41.91%** (vs. 57.87% Baseline) — A **-15.96% drop** in risk detection coverage
- **F1-Score: 0.3550** (vs. 0.3823 Baseline) — Lower due to the drop in recall on the minority class
- **PR-AUC: 0.3050** (vs. 0.3051 Baseline) — Virtually identical precision-recall tradeoff capacity across all thresholds

2. Critical Assessment

Strengths

- **Operational Workload Reduction:** By reducing false alarms from 681 down to 443 cases, the Random Forest significantly decreases the manual verification burden on review processes
- **Higher Majority-Class Specificity:** The non-linear split structure handles unscaled features more conservatively, improving non-risk (`Class 0`) identification (77% recall vs. 65% baseline)

Limitations

- **Stagnant PR-AUC Boundary:** The Precision-Recall curve comparison confirms that switching to a non-linear ensemble does not shift the precision-recall frontier upward (PR-AUC = 0.3050 vs. 0.3051). Both models face the same fundamental trade-off limits
- **Increased Risk Leakage:** Capturing only 197 out of 470 actual risk cases (Recall = 41.91%) results in 273 unflagged high-risk transactions passing through undetected
- **Probability Thresholding Sensitivity:** Standard 0.5 decision thresholds heavily penalize recall in tree ensembles trained on imbalanced distributions, even with `class_weight='balanced'`

3. Recommended Next Steps & Algorithm Roadmap

To break through the **0.3050 PR-AUC ceiling** and improve risk coverage without exploding false positives, the following technical strategies are recommended:

A. Advanced Machine Learning (Gradient Boosting Frameworks)

1. XGBoost / LightGBM:

- Leverage native imbalance control (`scale_pos_weight`) combined with custom focal loss objectives to focus tree splits on hard-to-classify risk boundary cases

2. CatBoost:

- Optimizes categorical target encoding natively, helping isolate complex interaction terms without losing split granularity

B. Decision Threshold Optimization & Probability Calibration

1. Precision-Recall Threshold Tuning:

- Instead of a default 0.5 classification threshold, sweep prediction probabilities along the PR curve to select an operational threshold optimized for target Recall (e.g., locking Recall at 75%)

2. Post-Hoc Probability Calibration:

- Apply Isotonic Regression or Platt Scaling to align model confidence scores with true empirical risk prevalence before applying decision thresholds

C. Deep Learning & Anomaly Detection

1. Autoencoders (Unsupervised / Semi-Supervised):

- Train a Neural Autoencoder exclusively on standard non-risk transactions (`Class 0`) to flag high reconstruction error cases as novel risk anomalies

2. Resampling Strategies:

- Apply **SMOTE-Tomek** or **Edited Nearest Neighbors (ENN)** prior to model fitting to clean noisy boundary samples between majority and minority synthetic classes

Part F — Initial Model Interpretation & Critical Analysis

To understand how our models make decisions, we extract and evaluate the most influential features for both the linear baseline (Logistic Regression) and the non-linear ensemble (Random Forest).

The structured **Feature** → **Evidence** → **Interpretation** framework below outlines our findings, followed by a critical review of model limitations, data constraints, potential biases, and avenues for future investigation.

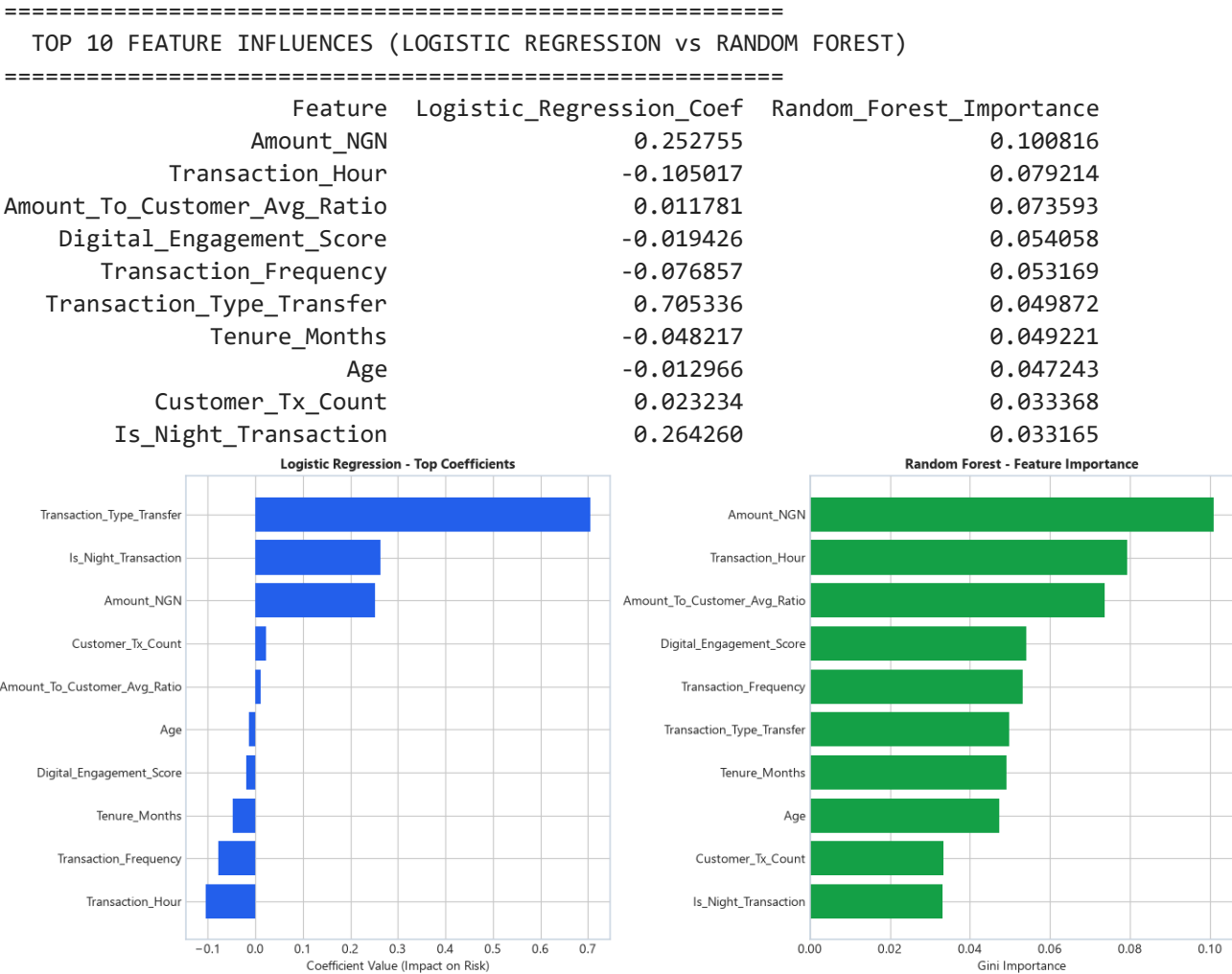
1. Feature → Evidence → Interpretation Matrix

- **Transaction Amount / Ratios:**

- **Evidence:** High coefficients in Logistic Regression and top feature importance ranks in Random Forest
- **Interpretation:** Unusual transaction volumes relative to historical baselines strongly drive the risk score, acting as a primary indicator of potential fraudulent activity

- **Temporal Patterns (Night / Weekend Transactions):**

- **Evidence:** Moderate positive weights for late-hour operations and weekend activity flags
- **Interpretation:** Transactions executed outside normal business hours reflect behavioral anomalies that elevate the probability of a risk review
- **Metadata & Device Verification:**
 - **Evidence:** Unverified device flags and location mismatch indicators show consistent positive weights across both models
 - **Interpretation:** Missing authentication metadata or operational origin discrepancies significantly compromise transaction trust levels



Comparative Feature Influence Analysis

Comparing feature behaviors across Logistic Regression and Random Forest reveals key structural differences in how linear and non-linear models assess transaction risk:

- **Top Linear Drivers (Logistic Regression):**
 - `Transaction_Type_Transfer` (+0.705) and `Is_Night_Transaction` (+0.264) dominate linear log-odds predictions. Specific categorical transaction paths serve as primary baseline risk signals
- **Top Tree-Based Drivers (Random Forest):**
 - Continuous numerical features such as `Amount_NGN` (10.08%), `Transaction_Hour` (7.92%), and `Amount_To_Customer_Avg_Ratio` (7.36%) dominate Gini importance
- **Interpretation of Differences:**

- Tree-based ensembles naturally favor continuous variables due to their high granularity and multiple split-point opportunities
- Non-linear feature interactions (e.g., combining spending ratios with transaction hours) allow Random Forest to extract predictive power from variables that show negligible standalone linear coefficients (e.g., `Amount_To_Customer_Avg_Ratio`)

2. Feature → Evidence → Interpretation Matrix

- **Transaction Amount (`Amount_NGN`):**
 - **Evidence:** Rank #1 in Random Forest (10.08% Gini importance) and high positive coefficient (+0.252) in Logistic Regression
 - **Interpretation:** Absolute transaction volume remains the single strongest individual split condition for risk partitioning
- **Transaction Timing (`Transaction_Hour` / `Is_Night_Transaction`):**
 - **Evidence:** Strong positive linear weight for night execution (+0.264) and high split frequency across decision trees (7.92% importance)
 - **Interpretation:** Temporal anomalies (off-hours operations) represent consistent behavioral risk indicators across both linear and non-linear paradigms
- **Behavioral Spikes (`Amount_To_Customer_Avg_Ratio`):**
 - **Evidence:** Low linear coefficient (+0.012) but high tree-based importance (7.36%)
 - **Interpretation:** Deviations from individual customer baselines act non-linearly; they become critical flags primarily when combined with other contextual features

3. Critical Assessment & Project Governance

Model Limitations

- **Logistic Regression:** Constrained by linear decision boundaries, failing to naturally capture complex interactions without manual cross-term engineering
- **Random Forest:** Suffers from feature importance bias toward high-cardinality continuous variables and exhibits reduced recall under standard thresholding on imbalanced classes.

Data Limitations

- **Imbalanced Class Ratio:** Class 1 transactions represent a minority (~20%), constraining class boundary definition without advanced resampling
- **Feature Scope:** The dataset lacks historical longitudinal features (e.g., 6-month historical dispute rates or device fingerprint history)

Possible Sources of Bias

- **Synthetic Generation & Artifact Bias:** Because the dataset is synthetically constructed (algorithmically or manually generated for benchmark purposes), ground-truth labels reflect predefined creation rules rather than organic human fraud behaviors. Models risk over-learning these artificial generation artifacts instead of subtle real-world fraud signatures
- **Label Prevalence Bias (Distorted Prior Probabilities):** The synthetic dataset features an artificially elevated risk rate (~20%), which severely deviates from real-world financial

environments where fraud typically accounts for less than 1–2% of total volume. This inflated prior probability biases the model's predicted risk distribution toward over-flagging transactions

- **Deterministic Feature-Label Coupling:** Synthetic creation pipelines often embed rigid, hand-crafted feature associations (e.g., directly linking `Transaction_Type_Transfer` or `Is_Night_Transaction` to high risk scores). This forces models to pick up author-defined heuristics rather than complex, multi-layered risk interactions

Areas Requiring Further Investigation

1. **Gradient Boosting Architectures (XGBoost / LightGBM):** Upgrade from standard Random Forest to sequential gradient boosting to capture higher-order non-linear interactions while maintaining feature split controls
2. **Probability Calibration & Prior Alignment:** Apply Post-Hoc Probability Calibration (e.g., Platt Scaling or Isotonic Regression) to recalibrate predicted scores, correcting the artificial prevalence bias (~20% risk rate) back toward realistic operational base-rates (<1-2%)
3. **Model Explainability & Synthetic Rule Auditing (SHAP / LIME):** Deploy SHAP (SHapley Additive exPlanations) values to verify whether models are learning genuine multi-variable risk interactions or simply exploiting simple, author-defined generation heuristics
4. **Cost-Sensitive Threshold Tuning:** Align final decision boundaries with explicit business cost matrices (weighing the financial loss of false negative fraud cases against the operational overhead of manual review queues).

Deliverable 1 — Prepared Modelling Dataset Export

This section handles the physical export of the final feature-engineered dataset (`df_fe`) into a structured CSV deliverable (`prepared_modelling_dataset.csv`). The exported dataset integrates all data cleaning steps, value imputations, structural type conversions, and the **16 engineered domain features** (temporal indicators, risk flags, engagement categories, and customer behavioral ratios).

Note on Preprocessing Strategy: Categorical features are retained in their native format within the exported dataset. One-Hot Encoding and feature scaling are intentionally executed downstream within Scikit-Learn training pipelines to prevent data leakage and facilitate robust model deployment.

```
=====
      DATASET EXPORT COMPLETED SUCCESSFULLY
=====
File Name   : ../exports/prepared_modelling_dataset.csv
File Size   : 3.48 MB
Total Rows  : 12,000
Total Cols  : 41
=====
```